INTERMEDIATE MICROECONOMICS · PROF. NAVEEN SUNDER

Week 2

Indifference Curves, Budget Constraints, and the Consumer Optimum

"The real price of everything is the toil and trouble of acquiring it."
— Adam Smith, The Wealth of Nations, 1776

- Why do airlines offer economy *and* business class?
- When fuel taxes rise, do consumers cut driving as predicted?
- Why might raising a price *increase* total revenue?
- How does a student rationally allocate a fixed weekly budget?
- Why do firms like Apple bundle features rather than price separately?
- When your salary doubles, does every category of spending double?

Learning Objectives and Topics

Each objective states what you will be able to *do*. The sections listed beneath it are where you learn to do it.

LO1 Identify and sketch the four IC types and explain the logic behind each shape.
§2.1 Recap *§2.2 Types of Goods* *§2.3 Substitutes* *§2.4 Complements* *§2.5 Neutral* *§2.6 Bads*

LO2 Derive the budget constraint; identify slope, intercepts, and the affordable set.
§2.7 Budget Constraint *§2.8 Plotting*

LO3 Predict and diagram how income changes (shift) and price changes (pivot) move the budget line.
§2.9 Changes in BC

LO4 State and apply the two conditions for an interior optimum: budget exhaustion and tangency.
§2.10 Optimal Bundle

LO5 Use the bang-for-the-buck condition to assess and correct a consumer's allocation.
§2.11 Bang for the Buck

LO6 Solve algebraically for the optimal bundle under Cobb-Douglas and perfect-complement preferences.
§2.12 Worked Examples

LO7 Select the correct optimisation method (tangency, corner, or kink) for any preference type.
§2.10 Decision Guide *§2.12 Worked Examples*

Notation Used in This Chapter.

Symbol	Meaning	Symbol	Meaning
X, Y	quantities of goods X and Y	$U(X, Y)$	utility function
P_x, P_y	prices of goods X and Y	MU_x	marginal utility of X
I	consumer income	MU_y	marginal utility of Y
X^*, Y^*	optimal quantities	MRS	marginal rate of substitution
IC	indifference curve	BC	budget constraint

2.1. Recap: What Is an Indifference Curve?

How do economists represent what a consumer wants, without asking them to put a dollar value on their happiness?

In the previous week, we introduced the concept of consumer preferences and indifference curves. To briefly recap: a consumer who faces two goods, X and Y, can be described by an **indifference curve** — a set of all bundles (X, Y) that provide the consumer with exactly the same level of utility (satisfaction). The consumer is, by definition, *indifferent* among all bundles on the same curve.

Several key properties hold for well-behaved preferences:

- Indifference curves are **downward sloping**: since both goods are desirable, giving up some of Y must be compensated by more of X to keep utility constant.
- They are **convex**: the consumer prefers variety, so they are willing to give up less and less of Y as they consume more X.
- They **do not cross**: two indifference curves representing different utility levels can never intersect.
- Moving **outward from the origin** (to the upper right) reaches higher utility levels — more of both goods is preferred.

This week, we examine what happens to these properties when the nature of the two goods changes. The shape of the indifference curve is not fixed — it depends entirely on the relationship between the two goods. We then introduce the second key ingredient of consumer theory: the **budget constraint**, which describes what the consumer can afford. Finally, we combine the two to find the consumer's **optimal bundle** — the best affordable choice.

Why This Matters. Indifference curves are the economist's way of representing preferences without asking people to assign dollar values to their satisfaction. Businesses use similar logic — through market research and conjoint analysis — to understand how customers trade off product features such as price, quality, and convenience. The same framework that describes a student choosing between food and clothing underlies how airlines design fare classes and

how phone manufacturers bundle features.

1. Name two properties that all well-behaved indifference curves share, and explain in one sentence why each property makes economic sense.
2. True or false: a consumer can be indifferent between a bundle with more of both goods and one with less of both goods. Justify your answer.

With the basic properties of indifference curves in place, we now ask: what happens to the shape *of the curve when the goods have a special relationship with each other? The next four sections answer this question.*

2.2. Types of Goods and Their Indifference Curves

The standard convex indifference curve is the right shape for a typical consumer who values both goods and prefers variety. But economics deals with many kinds of goods, and the shape of the indifference curve changes depending on the relationship between them. We examine four important cases.

Mini Case Study: A Student's Weekly Budget. Consider Maya, a college student with $60 per week to spend on coffee and textbooks. She values both, but they are clearly neither perfect substitutes (she cannot read a cup of coffee) nor perfect complements (she can enjoy coffee without a textbook in hand). Her preferences are well-behaved: she prefers more of each, she prefers variety, and her willingness to give up coffee for an extra textbook declines as she already has many textbooks. Her indifference curves are the standard convex shape — downward sloping and bowing toward the origin. The concepts in this section help explain how Maya's optimal choice changes if her budget grows or if textbook prices change.

2.3. Perfect Substitutes

If two products are truly identical in the eyes of the consumer, how does that change the shape of the indifference curve — and what does it mean for how a firm should price them?

2.3.1. Definition and Intuition

Perfect Substitutes: Two goods are **perfect substitutes** if the consumer regards them as identical — they serve the same function perfectly and the consumer is always willing to trade one for the other at a constant ratio.

The classic example is gasoline from two neighboring stations: Exxon gas and Mobil gas. For most consumers, a gallon of gasoline is a gallon of gasoline, regardless of brand. What matters is the total number of gallons purchased, not which station it comes from.

This has a direct implication for indifference curves. Consider a consumer starting at bundle $A = (2\text{ Mobil}, 2\text{ Exxon})$ — two gallons from each station, four gallons total.

- **Bundle** B = (3 Mobil, 1 Exxon): still four gallons total. Same utility as A.
- **Bundle** C = (4 Mobil, 0 Exxon): still four gallons total. Same utility as A.

Because the consumer only cares about the total quantity of gas and not the brand, all of these bundles lie on the same indifference curve. The willingness to trade one gallon of Exxon for one gallon of Mobil is constant — always one-for-one. This constant trade-off rate means the indifference curve is a **straight line** with slope -1.

2.3.2. The Shape: Straight Lines

More generally, if the consumer is always willing to trade a units of good Y for b units of good X, the indifference curve has slope $-a/b$. For the gas example, $a = b = 1$.

The utility function for perfect substitutes takes the linear form:

$$U(X, Y) = aX + bY$$

The MRS (the negative slope of the IC) is constant: MRS $= a/b$ everywhere.

> **Key Idea.** Because the consumer treats the two goods as identical, no amount of X or Y already consumed changes how willing they are to swap. The MRS is fixed. This is why the indifference curve is a straight line rather than the bowed shape we see for ordinary goods.

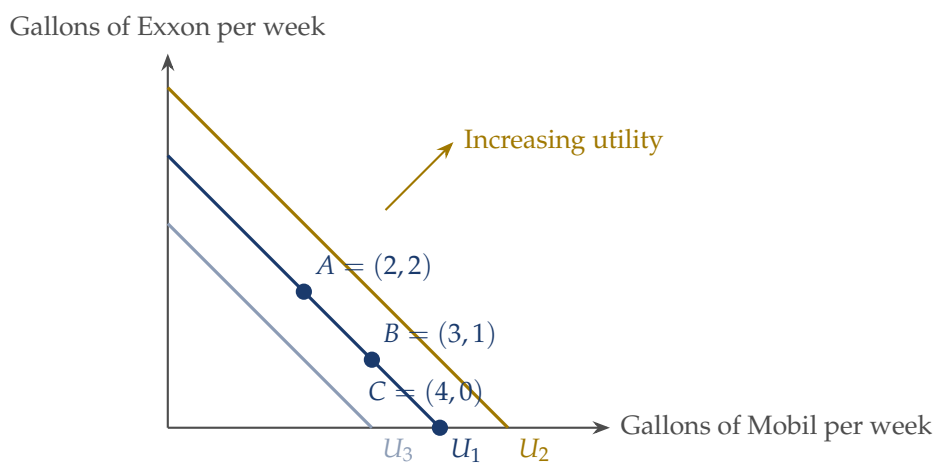


Figure 1. Perfect substitutes: Exxon vs. Mobil gasoline. Focus on two things: (1) all three curves are straight parallel lines — the hallmark of perfect substitutes; (2) bundles A, B, and C all lie on U_1 because each involves exactly 4 gallons total. The equal spacing between curves reflects the fact that each additional gallon raises utility by the same fixed amount regardless of brand mix.

> **Key feature of perfect substitutes:** The indifference curves are straight, parallel, downward-sloping lines. The MRS is constant — the consumer is always willing to trade the two goods at the same fixed rate, regardless of how much of each they already have.

> **Try It.** Suppose a consumer views Pepsi and Coke as perfect substitutes, always willing to trade 1 can of Pepsi for 1 can of Coke. Write down the utility function and the MRS. What does

the indifference curve look like? *(See Solution to Q2a.)*

Application 2.1: Store Brands and the Perfect-Substitute Threat. Retailers face perfect-substitute logic every time they launch a private-label product. If consumers genuinely view a store-brand paper towel as identical to the name-brand version, the indifference curve between the two is a straight line with slope -1. The consumer will always choose whichever is cheaper — the budget line will be tangent to the IC at a *corner*, buying only the cheaper brand. This is why Procter & Gamble and Unilever invest heavily in advertising: the goal is not just awareness but preference differentiation — shifting consumers away from a linear indifference curve toward a bowed one, so that they prefer a mix of the two brands rather than the cheapest-only corner solution.

1. A consumer has utility $U(X, Y) = 2X + 3Y$. What is the MRS? Draw one indifference curve for $U = 12$.
2. If two goods are perfect substitutes, will the consumer ever buy both simultaneously? Under what special condition might they?

Perfect substitutes give us straight-line indifference curves because the consumer is completely flexible about the mix. We now examine the opposite extreme: goods that must be consumed together in a rigid ratio.

2.3.3. Definition and Intuition

Perfect Complements: Two goods are **perfect complements** if they must be consumed together in fixed proportions in order to generate utility. Extra units of one good, without the corresponding extra units of the other, provide no additional utility.

The canonical example is **left shoes and right shoes**. A consumer with 2 left shoes and 2 right shoes has 2 functional pairs. Now consider the following alternatives:

- **Bundle** A = (2 left, 2 right): 2 functional pairs.
- **Bundle** B = (3 left, 2 right): still only 2 functional pairs. The extra left shoe is useless without a matching right shoe. Same utility as A.
- **Bundle** C = (4 left, 2 right): still 2 functional pairs. Same utility.
- If we instead increase right shoes: (2 left, 3 right), (2 left, 4 right) — still only 2 functional pairs. Same utility.

This means the indifference curve through $A = (2, 2)$ is **L-shaped**: a vertical arm at $X = 2$ (adding more right shoes when left shoes = 2 gives no extra utility) and a horizontal arm at $Y = 2$ (adding more left shoes when right shoes = 2 gives no extra utility). The kink is at the bundle $(2, 2)$.

More generally, for goods consumed in fixed proportions $a : b$, the utility is:

$$U(X, Y) = \min\left(\frac{X}{a}, \frac{Y}{b}\right)$$

For the shoe example with $a = b = 1$: $U = \min(X, Y)$.

2.3.4. The Shape: L-Shaped Curves

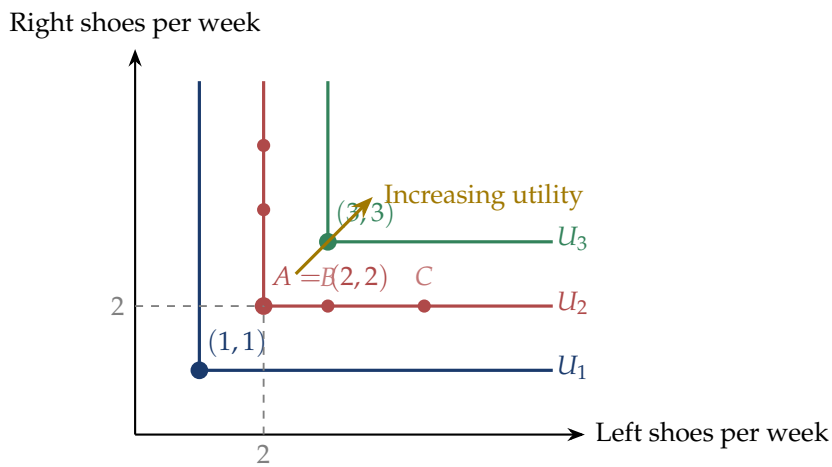


Figure 2. Perfect complements: left shoes and right shoes. Focus on the kink point of each curve — that is the only bundle where both goods are fully "used up." On the horizontal arm of U_2 (bundles like B and C), you have surplus left shoes that add nothing. On the vertical arm, surplus right shoes add nothing. Notice that all kinks lie on the 45° ray, confirming that the consumer always wants equal amounts of each good.

Key feature of perfect complements: The indifference curves are L-shaped with the kink on the ray $X/a = Y/b$ (for equal proportions, the 45° line). Extra units of one good, without the corresponding other good, provide zero marginal utility. The MRS is not defined at the kink, and is either 0 (on horizontal arm) or ∞ (on vertical arm).

Key Idea. The kink of the L is the only point on each indifference curve where both goods are being "used." Anywhere else on the L, one good is in excess — you have more of it than you can actually put to use. This is why extra units of the surplus good add no utility.

Application 2.2: The Razor-and-Blade Business Model. Perfect-complement preferences underlie one of the most successful pricing strategies in business history. Gillette sells razors cheaply (near cost) and charges a premium on replacement blades. Printer companies do the same with ink cartridges. Apple's ecosystem — iPhone and App Store — exhibits the same logic. When two goods are perfect complements, the consumer *must* buy them in a fixed ratio. A firm that controls supply of one complement can therefore extract value through the other. The indifference curve analysis here is the foundation for understanding why firms in technology, gaming (consoles and games), and healthcare (devices and consumables) adopt this two-part pricing structure.

1. A consumer has $U(X, Y) = \min(X, 2Y)$. Sketch two indifference curves and identify the kink points. Where do all kinks lie?
2. Why is the MRS undefined at the kink of an L-shaped indifference curve? What does this mean for applying the tangency condition?

We have now seen two extremes: goods that are fully interchangeable (substitutes) and goods that are rigidly tied together (complements). The next two sections examine goods where one of the two items is either irrelevant or actively harmful to the consumer.

Can a good have zero value to a consumer? What does the indifference map look like when one good simply does not matter?

2.3.5. Definition and Intuition

Neutral Good (Useless Good): A good is **neutral** if it provides zero marginal utility to the consumer — consuming more or less of it leaves the consumer's utility completely unchanged.

Examples include class notes for a course you have never taken, or a book in a language you do not speak. These are goods that, for this particular consumer, simply do not matter.

Now suppose good Y is the neutral good and good X is a normal good that the consumer values. What does the indifference curve look like?

Start at bundle $A = (2, 2)$ — 2 units of food (X) and 2 units of the neutral good (Y). To find another bundle with the same utility:

- Moving to $(2, 3)$: we added one unit of Y. But Y is neutral — it changes nothing. Utility is the same.
- Moving to $(2, 10)$: Y is still neutral. Utility is still the same.
- Any bundle of the form $(2, Y)$ for any $Y \geq 0$ has the same utility as A.

The indifference curve through A is therefore a **vertical line** at $X = 2$: the consumer is indifferent among all bundles with 2 units of food, regardless of the amount of the neutral good. Utility only changes when X changes, so utility increases as you move **horizontally to the right** (more of the valued good X).

1. For the utility function $U(X, Y) = X$, verify that Y is neutral by computing $\partial U / \partial Y$.
2. A consumer has $U(X, Y) = X + 0$. If their income doubles, does their demand for Y change? Explain without algebra.

A neutral good simply does not enter the consumer's utility function. An economic bad goes one

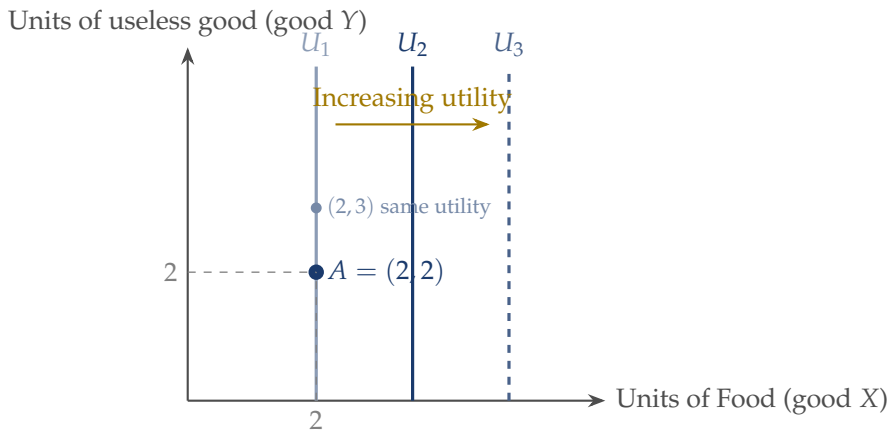


Figure 3. Neutral (useless) good: vertical indifference curves. Focus on the vertical lines and the rightward arrow. Point $A = (2, 2)$ and the point $(2, 3)$ lie on the *same* indifference curve U_1 — adding more of the neutral good Y changes nothing. Utility increases only when X increases, which is why higher ICs lie to the *right*, not above.

step further: it actively reduces utility. This gives rise to the one preference type with an upward-*sloping indifference curve.*

2.4. Economic Bads

How do economists model goods that consumers would prefer to avoid entirely — like pollution, noise, or a long commute?

2.4.1. Definition and Intuition

> **Economic Bad:** A good is an **economic bad** (or simply a "bad") if consuming more of it *reduces* the consumer's utility. It confers negative marginal utility. The more formal term is **disutility**.

Classic examples include pollution, noise, or commute time — things consumers must endure but which make them worse off.

Suppose good Y is an economic bad (e.g., air pollution) and good X is a normal good (e.g., food). Starting at bundle $A = (2, 2)$:

- If we add more Y (more pollution), the consumer is worse off.
- To compensate and keep utility constant, we must give them more X as well.
- Therefore, moving along the IC requires us to move to the upper right: more X to compensate for more Y.

The indifference curve through A is therefore **upward sloping**: as Y (the bad) increases, X must also increase to keep the consumer at the same utility level. Higher utility is reached by moving **to the right** (more X) or **downward** (less of the bad Y).

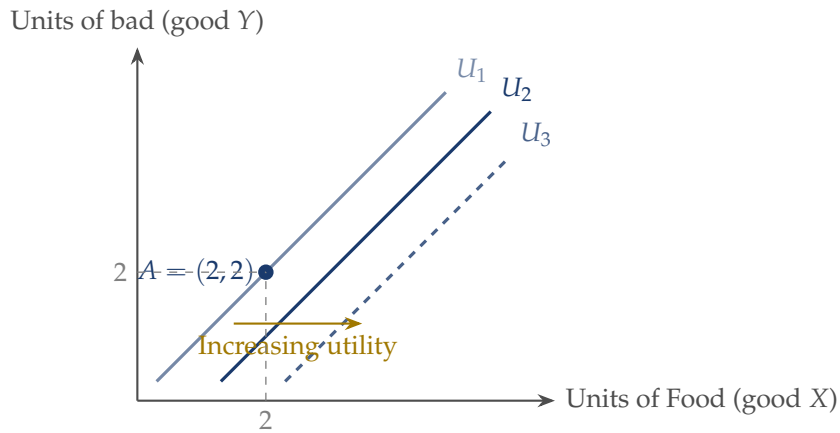


Figure 4. Economic bad: upward-sloping indifference curves. Focus on two features: (1) all three curves slope *upward* — the exact opposite of ordinary goods; (2) the arrow shows that higher utility is reached by moving *right* (more food, same pollution) not upward. Bundle $A = (2, 2)$ lies on U_1. To stay on U_1 when pollution Y rises above 2, the consumer must receive more food X as compensation — hence the upward slope. Moving right to U_2 or U_3 requires more food but no extra pollution.

Why This Matters. Economic bads appear constantly in policy debates. When a government evaluates a highway expansion, residents near the road experience noise and pollution as economic bads. To keep utility constant, they need compensation — more of some other good (income, parks, soundproofing). Understanding indifference curves for bads helps economists design sensible compensation policies and value environmental amenities.

1. Commute time is an economic bad for most workers. If a firm offers a pay rise to compensate employees for a longer commute, what does this say about the shape of their indifference curves over income and commute time?
2. For the utility function $U(X, Y) = X - Y^2$ (where $Y \geq 0$), show that Y is a bad by computing $\partial U / \partial Y$.

Summary of IC shapes by good type:

Good type	IC shape
Standard (both goods desired)	Downward-sloping, convex
Perfect substitutes	Straight line, downward sloping
Perfect complements	L-shaped
Neutral good (Y useless)	Vertical line
Economic bad (Y bad)	Upward sloping

We now have a complete picture of what consumers want. *But preferences alone do not determine choices — a consumer also faces a hard limit on what they can afford. We turn to that constraint now.*

2.5. The Budget Constraint

A student has $60 a week for coffee and textbooks. What is the full set of bundles she can afford, and how does that set change if textbook prices rise?

Indifference curves describe what the consumer *wants*. But wanting something is not enough — the consumer also faces a resource constraint. This is where the **budget constraint** (BC) comes in.

Budget constraint: the set of all bundles the consumer can afford given income and prices.

2.5.1. The Budget Constraint Equation

Suppose a consumer has income I and faces prices P_x for good X and P_y for good Y. If the consumer spends all of their income on these two goods:

- Amount spent on good X: $P_x \cdot X$ (price times quantity).
- Amount spent on good Y: $P_y \cdot Y$.
- Total spending = income: $I = P_x \cdot X + P_y \cdot Y$.

The equation below simply says that all income goes to one of two uses — buying X or buying Y. The **budget constraint** is the equation:

$$\boxed{I = P_x \cdot X + P_y \cdot Y} \tag{1}$$

I, P_x, and P_y are all **numbers** — they are given to the consumer. The unknowns are the quantities X and Y. The budget constraint is a **linear equation** in X and Y — it describes a straight line when plotted.

> **Key Idea.** The budget constraint separates the feasible from the infeasible. Any bundle on or below the line is affordable; any bundle above the line requires more money than the consumer has. The line itself represents all the ways the consumer can spend every last dollar.

2.5.2. The Intercepts and Slope

Rearranging the budget constraint so that Y is on the left gives us the slope-intercept form. This makes it easy to plot and to read off the key values:

$$Y = \frac{I}{P_y} - \frac{P_x}{P_y} X \tag{2}$$

From this form, we can read off the key properties:

- **Y-intercept** ($X = 0$): $Y_{\text{max}} = \dfrac{I}{P_y}$.

  This is the maximum amount of good Y the consumer can afford if they spend all income on Y.

- **X-intercept** ($Y = 0$): $X_{\text{max}} = \dfrac{I}{P_x}$.

This is the maximum amount of good X the consumer can afford if they spend all income on X.

- **Slope** $= -\dfrac{P_x}{P_y}$.

  The slope is negative (the budget line is downward sloping): buying more X requires giving up some Y. The slope measures the **opportunity cost** of X in terms of Y — for every extra unit of X purchased, the consumer must give up P_x/P_y units of Y.

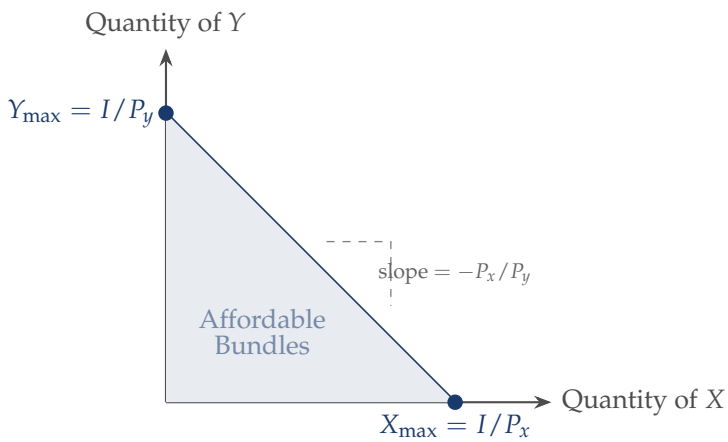


Figure 5. The budget constraint. Focus on the two intercepts: the Y-intercept is the most Y the consumer can buy (spending everything on Y); the X-intercept is the most X they can buy. The slope $-P_x/P_y$ tells you exactly how many units of Y must be sacrificed for one more unit of X. Every point in the shaded triangle is affordable; every point above the line is not.

Worked Example: Sheela's Budget Constraint.

Sheela has income $I = \$100$, with $P_x = \$5$ and $P_y = \$2$.

Step 1 — Maximum quantities:

$$X_{\text{max}} = \frac{I}{P_x} = \frac{100}{5} = 20, \qquad Y_{\text{max}} = \frac{I}{P_y} = \frac{100}{2} = 50$$

Step 2 — Budget constraint equation:

$$100 = 5X + 2Y$$

Step 3 — Slope:

$$\text{Slope} = -\frac{P_x}{P_y} = -\frac{5}{2} = -2.5$$

In words: For every additional unit of food Sheela buys, she must give up 2.5 units of clothing.

Step 4 — Draw the line: Plot $(0, 50)$ on the Y-axis and $(20, 0)$ on the X-axis; connect them with a straight line.

Try It. Suppose Sheela's income rises to \$150 but prices stay the same. Without calculating, predict what happens to the budget line. Then verify by computing the new intercepts. *(Answer: both intercepts scale by 1.5; slope unchanged.)*

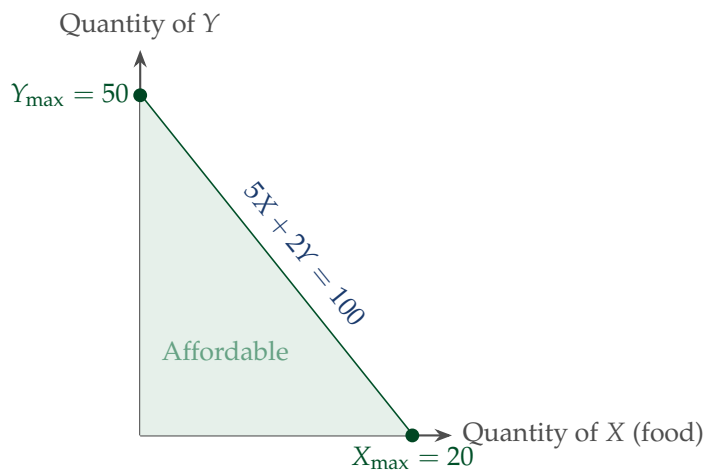


Figure 6. Sheela's budget constraint: $5X + 2Y = 100$**.** Notice that the budget line is pinned by exactly two numbers: $Y_{\max} = 50$ on the vertical axis and $X_{\max} = 20$ on the horizontal axis. Any bundle inside the shaded triangle costs less than \$100; any bundle on the line costs exactly \$100.

1. A consumer has $I = \$120$, $P_x = \$4$, $P_y = \$6$. Write the budget constraint, find both intercepts, and state the slope. What is the opportunity cost of one unit of X?
2. True or false: if P_x and P_y both double while income stays constant, the budget line shifts inward in parallel. Justify carefully.

We now know how to draw the budget constraint for a given income and set of prices. The next question is: how does the line move when those numbers change?

2.6. Changes in the Budget Constraint

When a government cuts fuel taxes, does the budget line shift or rotate? Getting this right is the foundation for predicting how consumers respond to price changes.

The budget constraint is not fixed. It shifts whenever income or prices change. Understanding how it shifts is essential for understanding how consumer choices change.

2.6.1. Income Increases: Parallel Outward Shift

How does the BC change if income increases from \$100 to \$150, while $P_x = 5$ and $P_y = 2$ remain the same?

Think about what changes and what stays the same:

- Y**-intercept** $= I/P_y$: increases from $100/2 = 50$ to $150/2 = 75$.
- X**-intercept** $= I/P_x$: increases from $100/5 = 20$ to $150/5 = 30$.
- **Slope** $= -P_x/P_y = -5/2$: **unchanged** (prices did not change).

Both intercepts increase proportionally, and the slope is unchanged. The budget line **shifts outward in parallel** — it moves away from the origin without rotating. This makes sense: higher income means the consumer can afford more of everything, but the relative prices (and hence the trade-off between goods) are unchanged.

> **Key Idea.** Income shifts the budget constraint parallelly. The slope is set by prices alone, so as long as prices do not change, a higher income simply slides the entire line outward. Think of the line as a conveyor belt that moves north-east but keeps the same angle.

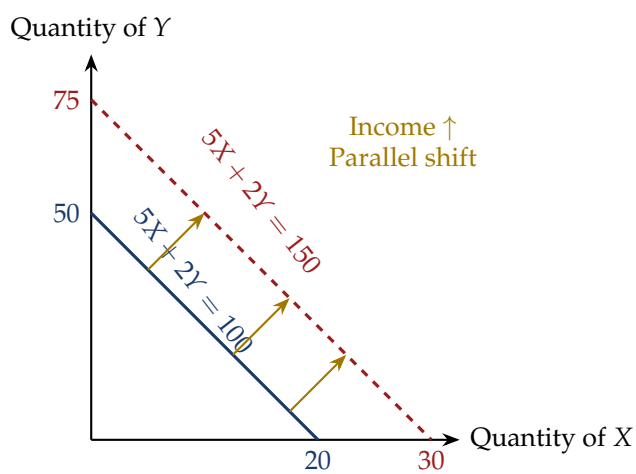


Figure 7. Income increase: parallel outward shift. The key observation is that the two lines are *parallel* — same slope, different intercepts. Both intercepts move by the same proportion (50%), confirming that only income changed. If you see a shift where the slope also changes, income alone cannot be the cause.

> **Income change ⇒ parallel shift of budget line.** The slope $(-P_x/P_y)$ is unchanged because prices have not moved. Both intercepts change proportionally with income.
>
> - Income **increases**: BC shifts **outward** (away from origin).
> - Income **decreases**: BC shifts **inward** (toward origin).

2.6.2. Price Change: Pivoting the Budget Line

How does the BC change if P_x decreases from \$5 to \$4, while income = \$100 and P_y = \$2 remain the same?

- Y**-intercept** $= I/P_y = 100/2 = 50$: **unchanged** (neither I nor P_y changed).
- X**-intercept** $= I/P_x$: increases from $100/5 = 20$ to $100/4 = 25$.
- **Slope** $= -P_x/P_y$: changes from $-5/2 = -2.5$ to $-4/2 = -2$ (becomes *flatter*).

When P_x falls:

- The Y-intercept is fixed (the consumer can still buy the same maximum of Y if they spend everything on Y).
- The X-intercept moves outward (the consumer can now buy more of X with all income).
- The line **pivots outward** around the fixed Y-intercept.

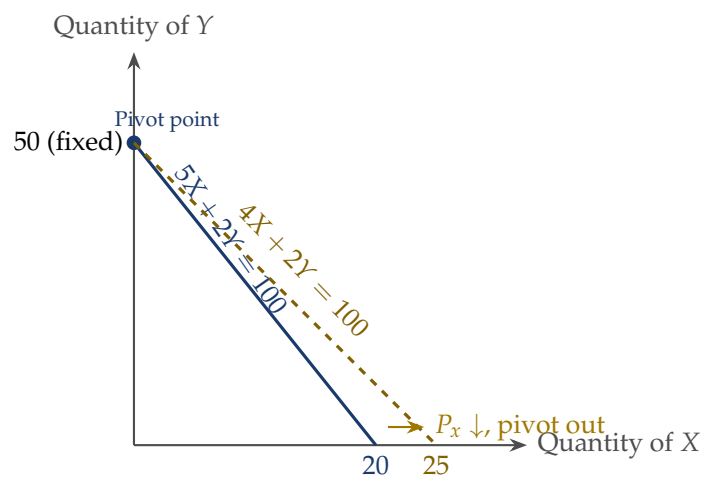


Figure 8. Price decrease: pivot around the fixed intercept. Notice that both lines share the same Y-intercept at 50 — this is the "pivot point." Only the X-intercept changes (from 20 to 25), and the line becomes flatter. Whenever you see one intercept fixed and the other moving, a price change is the cause.

Own-price change ⇒ pivot of budget line. The intercept on the *other* good's axis is fixed; only the intercept on the good whose price changed moves.

- P_x **falls**: BC pivots **outward** (right) around the Y-intercept. Line becomes flatter.
- P_x **rises**: BC pivots **inward** (left) around the Y-intercept. Line becomes steeper.

Common Mistake: Confusing Shifts with Pivots. Students often draw a parallel shift when a price changes, or a pivot when income changes. The rule is simple: *income changes both intercepts proportionally and leaves the slope unchanged* (parallel shift); *a price change fixes one intercept and moves the other, changing the slope* (pivot). Quick check: if the fixed intercept has moved, it is not a pivot.

Quick-Reference: What Moves on the Budget Line?

Change	Y-intercept	X-intercept	Slope	Type
Income ↑	Moves out	Moves out	Fixed	Parallel shift out
Income ↓	Moves in	Moves in	Fixed	Parallel shift in
$P_x \downarrow$	Fixed	Moves out	Flatter	Pivot out
$P_x \uparrow$	Fixed	Moves in	Steeper	Pivot in
$P_y \downarrow$	Moves out	Fixed	Steeper	Pivot out
$P_y \uparrow$	Moves in	Fixed	Flatter	Pivot in
All prices $\times k$	Moves in	Moves in	Fixed	$\equiv$ income $\div k$

Why This Matters. Every time a government raises or lowers a sales tax on a specific good, it effectively pivots consumers' budget lines. When gasoline taxes rise, the budget line pivots inward around the other good's intercept, and consumers demand less gasoline. Understanding this geometry is essential for predicting how tax policy changes behavior — a core task for both government economists and corporate strategists.

Application 2.3: Fuel Tax vs. Income Tax — Same Revenue, Different Behaviour. Suppose the government needs to raise \$500 per household in revenue. It can do so either by levying a tax on petrol (pivoting the budget line inward around the non-petrol intercept) or by reducing all households' income by \$500 (a parallel shift inward). Both policies raise the same revenue. But they affect consumer choices very differently: the fuel tax changes relative prices and discourages petrol consumption specifically; the income tax leaves relative prices intact and reduces consumption of all goods proportionally. This is why environmental economists prefer fuel taxes to income taxes when they want to reduce emissions — the pivot, not the shift, does the work.

1. Sheela has $I = \$100$, $P_x = \$5$, $P_y = \$2$. Draw the original budget line. Now P_y rises to \$4. Which intercept is fixed? Redraw and state the new slope.
2. A student claims: "doubling all prices is the same as halving my income." Is she correct? Use the budget constraint equation to verify.

We now know precisely how the budget line moves in response to any change in income or prices. We are ready to put the two pieces together — indifference curves and the budget constraint — to find the consumer's best choice.

2.7. Determining the Optimal Consumption Bundle

Given that a consumer wants to reach the highest indifference curve and can only afford bundles on or below the budget line, where exactly is the best bundle — and how do we find it?

2.7.1. The Central Question

We now have two pieces: (1) indifference curves, which describe what the consumer *wants*, and (2) the budget constraint, which describes what they can *afford*. The consumer's goal is:

Choose the bundle that provides the highest utility while remaining within the budget.

Intuition first. Imagine sliding an indifference curve outward from the origin — toward higher and higher utility levels. At some point, the curve just barely touches the budget line. Move it any further and it lies entirely above the line — unaffordable. The point of contact is the optimum. This geometric insight tells us everything: the optimal bundle is where the budget line is **tangent** to an indifference curve.

In terms of the diagram: find the indifference curve that is furthest from the origin (highest utility) while still having at least one point that lies on or inside the budget constraint.

2.7.2. Why Interior Points Are Not Optimal

Point A: interior point (inside the budget set).

Consider a bundle A that lies strictly inside the budget constraint (below and to the left of the line). The consumer is spending less than their income — some income is unspent. But since both goods are desirable, spending that remaining income on more X or Y would push the consumer to a higher indifference curve. Therefore, A is not optimal.

Conclusion: The optimal bundle must lie on the budget line — the consumer should spend all income.

Point B: on the budget line but not at a tangency.

Now consider bundle B on the budget line, where an indifference curve *crosses* (rather than is tangent to) the budget line. At a crossing, the IC intersects the budget line at two points. Between those two points on the budget line, there are bundles that lie above the IC through B and are affordable. This means the consumer can do better than B by moving along the budget line. Therefore, B is not optimal either.

Point C: tangency point.

Point C is where the budget line is *tangent* to an indifference curve. At this point:

1. The consumer is spending all of their income (on the budget line).
2. There is no affordable bundle that lies on a higher indifference curve — the tangency ensures that any deviation along the budget line moves to a *lower* IC.

Point C is the optimal consumption bundle.

2.7.3. The Two Conditions for Optimality

From the geometric analysis, two conditions must hold at the optimal bundle:

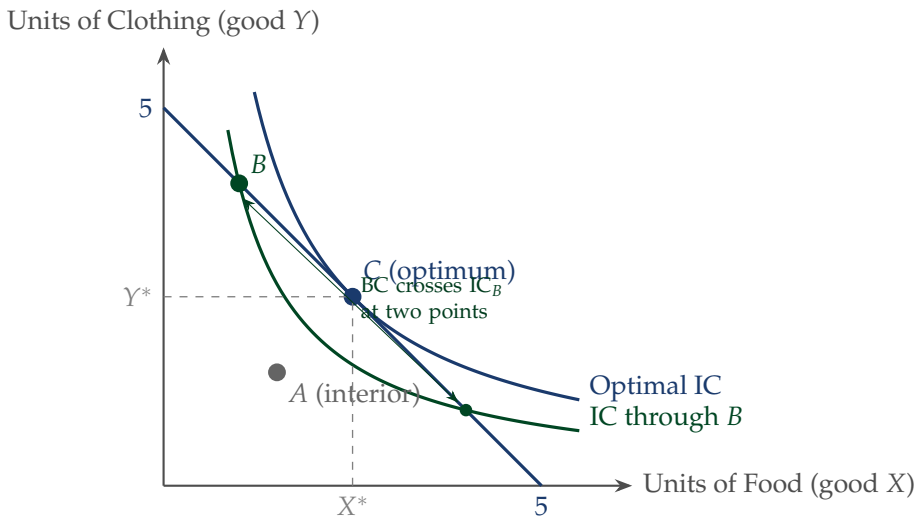


Figure 9. Finding the optimal bundle: why only C works. Look at all three points carefully. Point A (grey) lies strictly inside the budget set — income is being wasted. Point B (green) is on the budget line, but its indifference curve *crosses* the line at two points (marked), so there are affordable bundles on a higher IC. Only at C (blue) does the budget line *touch* the optimal IC without crossing it — the tangency guarantees no affordable improvement is possible. Note that $C = (2.5, 2.5)$ lies exactly on both the budget line ($y = 5 - x$) and the optimal IC ($y = 6.25/x$).

1. **Budget exhausted:** The consumer spends all income.

$$P_x \cdot X^* + P_y \cdot Y^* = I \tag{BC}$$

In words: Every dollar of income is allocated to one of the two goods — none is left unspent.

2. **Tangency condition:** The slope of the budget constraint equals the slope of the indifference curve at the optimal bundle.

$$\underbrace{-\frac{P_x}{P_y}}_{\text{slope of BC}} = \underbrace{-\frac{MU_x}{MU_y}}_{\text{slope of IC (MRS)}}$$

Cancelling the negative signs:

$$\boxed{\frac{P_x}{P_y} = \frac{MU_x}{MU_y}} \tag{3}$$

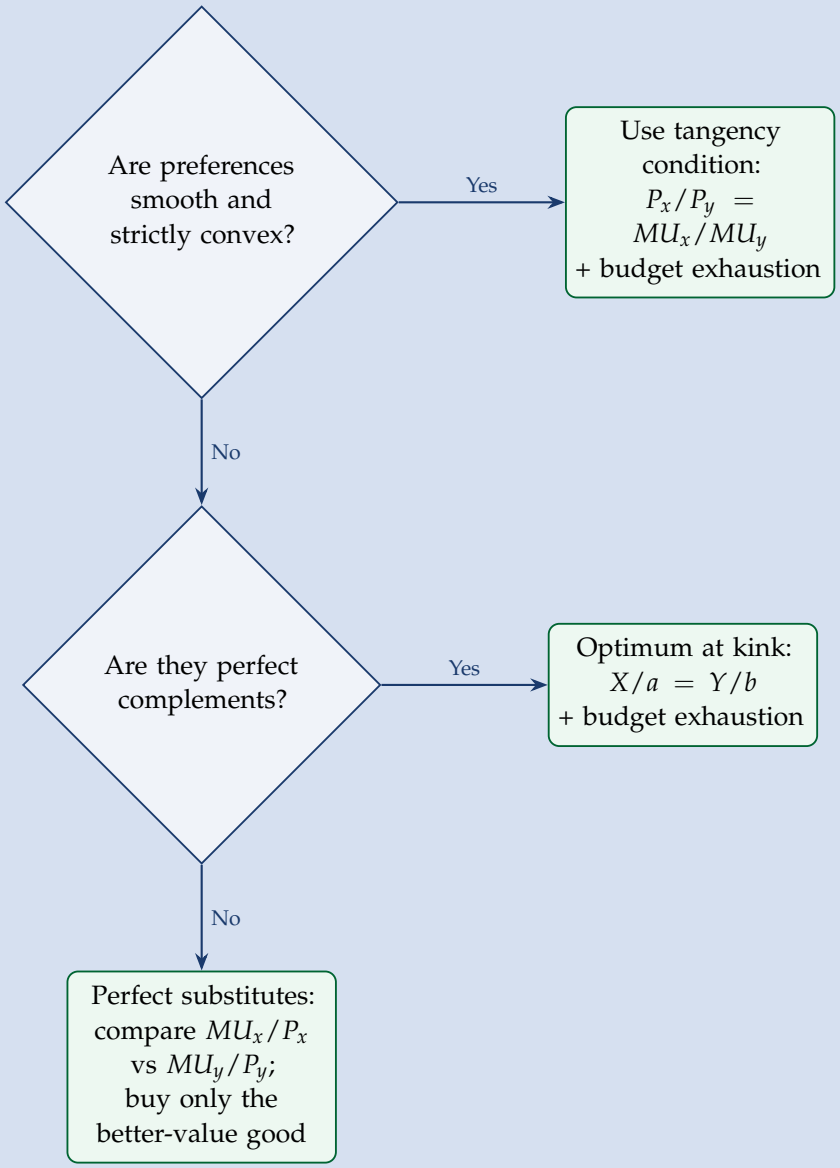
In words: The market's trade-off between the two goods (the price ratio) must equal the consumer's personal trade-off (the MRS). If they differ, the consumer can always rearrange spending to reach a higher indifference curve.

Key Idea. The tangency condition (3) says: at the optimum, the market's exchange rate between X and Y (the price ratio) exactly equals the consumer's personal exchange rate (the MRS). If they differed, the consumer could rearrange spending to get more utility without

spending more money.

Common Mistake: Using Tangency for All Preference Types. The tangency condition holds only for smooth, strictly convex indifference curves (like Cobb-Douglas). For perfect complements, the optimum is at the kink of the L, where the MRS is undefined. For perfect substitutes, the optimum is typically at a corner. Always check the type of preferences before applying the tangency condition mechanically.

Which Method Do I Use? A Decision Guide.



Check Your Understanding

1. A consumer is at a bundle where $MU_x/P_x > MU_y/P_y$ and the budget is exhausted. Are they at the optimum? If not, how should they adjust?

2. Explain in one sentence why a rational consumer with positive marginal utilities for both goods will never choose a bundle strictly inside the budget constraint.

We have established the two conditions for an optimal bundle. The next section reinterprets the tangency condition in a way that is deeply intuitive and immediately applicable to business decisions.

The optimality condition can be rearranged to yield a very intuitive interpretation. Rearranging $\frac{P_x}{P_y} = \frac{MU_x}{MU_y}$:

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \tag{4}$$

What does each side mean?

- MU_x is the extra utility from consuming one more unit of X — the **benefit** of X.
- P_x is the cost of one unit of X — the **cost** of X.
- Therefore, MU_x/P_x is the "utility per dollar" spent on X — the **benefit-to-cost ratio** for good X, sometimes called the "bang for the buck" from X.

The optimality condition says: *at the optimal bundle, the utility per dollar spent is the same for every good.* If this were not so — say, $MU_x/P_x > MU_y/P_y$ — then the consumer could increase total utility by spending one less dollar on Y and one more dollar on X, contradicting optimality.

Bang for the Buck: At the optimal consumption bundle,

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

The "utility per dollar" must be equalized across all goods. If it is higher for good X, the consumer should reallocate spending toward X.

Why This Matters. The bang-for-the-buck condition applies far beyond individual consumers. A firm deciding how to allocate a marketing budget across TV and digital advertising should equalize the return per dollar across channels. A hospital system allocating resources across departments, or a government allocating spending across programs, faces the same logic. Wherever resources are scarce and alternatives exist, equating marginal return per dollar is the path to efficiency.

Application 2.4: Marketing Budget Allocation. A coffee shop owner has \$10,000 to split between espresso machine upgrades and barista training. The last \$1,000 spent on machines increases daily throughput (and thus revenue) by \$80. The last \$1,000 spent on training improves customer satisfaction scores, increasing repeat visits worth \$50 per week — about \$130 per \$1,000 over a quarter. Since the return per dollar is higher for training (\$130) than for ma-

chines ($80), the owner should shift spending toward training. She keeps reallocating until the marginal returns per dollar are equalized. This is the bang-for-the-buck condition applied directly to a capital budgeting decision — the same logic that governs how Netflix allocates development resources across content categories or how a hospital system distributes funds across departments.

2.8. Finding the Optimal Bundle: Worked Examples

How do we translate the two conditions for optimality into a concrete step-by-step calculation?

Worked Example 1: Amanda's Optimal Consumption Bundle

Amanda has income $I = \$200$. The price of food ($X$) is $P_x = \$4$ and the price of clothing (Y) is $P_y = \$5$. Her utility function is $U(X, Y) = \sqrt{XY} = X^{1/2}Y^{1/2}$.

Find: (a) the budget line and slope; (b) the marginal utilities; (c) the MRS; (d) the optimal bundle; (e) the optimal bundle if P_x changes to $2.

Solution:

(a) Budget line and slope.

The budget constraint is:

$$I = P_x X + P_y Y \implies 200 = 4X + 5Y$$

Slope $= -P_x/P_y = -4/5$.

(b) Marginal utilities.

$U = X^{1/2}Y^{1/2}$.

$$MU_x = \frac{\partial U}{\partial X} = \frac{1}{2}X^{-1/2}Y^{1/2} = \frac{1}{2}\sqrt{\frac{Y}{X}}$$

$$MU_y = \frac{\partial U}{\partial Y} = \frac{1}{2}X^{1/2}Y^{-1/2} = \frac{1}{2}\sqrt{\frac{X}{Y}}$$

(c) Marginal Rate of Substitution (slope of IC).

The MRS is the ratio of marginal utilities, which simplifies to Y/X for this utility function:

$$\text{MRS} = -\frac{MU_x}{MU_y} = -\frac{\frac{1}{2}\sqrt{Y/X}}{\frac{1}{2}\sqrt{X/Y}} = -\frac{Y}{X}$$

So the slope of the indifference curve at any bundle (X, Y) is $-Y/X$.

(d) Optimal bundle (with $P_x = 4$, $P_y = 5$).

Two conditions must hold:

Condition 1 (budget exhausted):

$$200 = 4X + 5Y \tag{1}$$

Condition 2 (tangency: slope of BC = slope of IC):

$$-\frac{P_x}{P_y} = -\frac{Y}{X} \implies \frac{4}{5} = \frac{Y}{X} \implies Y = \frac{4X}{5} \tag{2}$$

Substitute (2) into (1):

$$200 = 4X + 5 \cdot \frac{4X}{5} = 4X + 4X = 8X \implies X^* = 25$$

$$Y^* = \frac{4 \times 25}{5} = 20$$

Optimal bundle: $(X^*, Y^*) = (25, 20)$**.**

Verification: $4(25) + 5(20) = 100 + 100 = 200 = I$. ✓

(e) Optimal bundle with $P_x = \$2$ (price of food changes).

Only the price of X changes. The new budget constraint is:

$$200 = 2X + 5Y \tag{1'}$$

The tangency condition now uses the new P_x:

$$\frac{P_x}{P_y} = \frac{Y}{X} \implies \frac{2}{5} = \frac{Y}{X} \implies Y = \frac{2X}{5} \tag{2'}$$

Substitute (2′) into (1′):

$$200 = 2X + 5 \cdot \frac{2X}{5} = 2X + 2X = 4X \implies X^* = 50$$

$$Y^* = \frac{2 \times 50}{5} = 20$$

New optimal bundle: $(X^*, Y^*) = (50, 20)$**.**

Observation: When the price of food fell from \$4 to \$2, Amanda doubled her food consumption from 25 to 50. Her clothing consumption stayed at 20. This makes sense: for this utility function ($\sqrt{XY}$), exactly half of income is always spent on each good. So clothing spending $= 5 \times 20 = \$100 = I/2$ in both cases.

> **Economic Insight.** When two goods are consumed under Cobb-Douglas preferences, a fall in the price of one good raises its consumption but leaves consumption of the other good completely unchanged. The freed-up spending goes entirely into the cheaper good — a result that turns out to be useful for understanding demand elasticities in Chapter 3.

Try It. In the Amanda example, verify that the bang-for-the-buck condition holds at the optimal bundle $(25, 20)$ with $P_x = 4$, $P_y = 5$. Compute MU_x/P_x and MU_y/P_y and check they are equal. *(See solution to Q10b.)*

Reverse Problem 1: Recovering a Price from an Observed Bundle.

Amanda consumes the bundle $(X^*, Y^*) = (40, 16)$ when her income is $I = \$200$ and $P_y = \$5$. Her utility function is $U(X, Y) = X^{1/2}Y^{1/2}$. What must P_x be?

Step 1 — Use budget exhaustion:

$$P_x \cdot 40 + 5 \cdot 16 = 200 \implies 40P_x = 200 - 80 = 120 \implies P_x = 3$$

Step 2 — Verify tangency condition:

$$\frac{P_x}{P_y} = \frac{3}{5} \qquad \text{and} \qquad \frac{MU_x}{MU_y} = \frac{Y}{X} = \frac{16}{40} = \frac{2}{5}$$

These are not equal, which tells us the bundle $(40, 16)$ is *not* a valid optimum at $P_x = 3$. The correct optimum at $P_x = 3$ uses the tangency condition: $Y/X = 3/5$, so $Y = 3X/5$. Substituting: $3X + 5(3X/5) = 200 \implies 6X = 200 \implies X^* = 33.3$, $Y^* = 20$.

Economic Insight: The reverse problem reveals whether an observed consumption bundle is consistent with rational optimisation. If budget exhaustion holds but the tangency condition fails, the consumer is not maximising utility at those prices.

Worked Example 2: Cobb-Douglas Demand Functions (General Case)

Generalizing Amanda's problem: suppose $U(X, Y) = X^\alpha Y^\beta$ and the budget constraint is $P_x X + P_y Y = I$. Find the general demand functions.

Solution:

$MU_x = \alpha X^{\alpha-1}Y^\beta$; $MU_y = \beta X^\alpha Y^{\beta-1}$.

Tangency condition: $\dfrac{MU_x}{MU_y} = \dfrac{P_x}{P_y}$:

$$\frac{\alpha Y}{\beta X} = \frac{P_x}{P_y} \implies Y = \frac{\beta P_x}{\alpha P_y} X$$

Substituting into the budget constraint:

$$P_x X + P_y \cdot \frac{\beta P_x}{\alpha P_y} X = I \implies P_x X\left(1 + \frac{\beta}{\alpha}\right) = I \implies P_x X \cdot \frac{\alpha + \beta}{\alpha} = I$$

$$\boxed{X^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{I}{P_x}, \qquad Y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{I}{P_y}}$$

Interpretation: the consumer spends fraction $\alpha/(\alpha + \beta)$ of income on X and fraction $\beta/(\alpha + \beta)$ on Y. For Amanda ($\alpha = \beta = 1/2$): $X^* = I/(2P_x)$, $Y^* = I/(2P_y)$ — exactly half of income on each, which is consistent with the numerical solution.

Economic Insight. The fixed-expenditure-share property of Cobb-Douglas preferences is one of the most empirically useful results in consumer theory. When a government or market research firm reports that "households spend roughly 30% of income on housing regardless of income level," they are observing Cobb-Douglas behaviour. This result will reappear in Chapter 3 when we compute income and price elasticities of demand.

Why This Matters. The Cobb-Douglas demand result — that the spending share on each good is fixed regardless of prices and income — is a workhorse of applied economics. It underlies many macroeconomic models and is used in practice to calibrate how households split income between categories like food, housing, and services.

The worked examples above have shown you the mechanics. The practice questions that follow are designed to test whether you can apply those mechanics from different angles — both forward (finding the optimum given preferences and prices) and in reverse (recovering parameters from observed behaviour). Attempt every question before consulting the solutions.

Where Does This Take Us?

- **Chapter 3** builds directly on the tangency condition to derive the consumer's individual demand curve — showing how the optimal bundle changes as price changes.
- The **Cobb-Douglas demand shares** derived here will reappear as the foundation for computing income and substitution effects.
- **Chapter 4** uses the budget constraint framework to measure consumer surplus and evaluate welfare effects of price changes and taxes.
- **Production theory (Chapter 6)** uses an identical apparatus — isoquants and isocost lines play exactly the roles that indifference curves and budget lines play here.

Five Common Exam Mistakes — and How to Avoid Them

1. **Drawing a parallel shift when a price changes.** A price change pivots the budget line around one fixed intercept — it never shifts both intercepts equally. Only an income change produces a parallel shift.
2. **Applying the tangency condition to perfect complements or substitutes.** The tangency condition requires a smooth, convex IC. For perfect complements, use the kink condition $X/a = Y/b$. For perfect substitutes, compare MU_x/P_x vs MU_y/P_y and buy only the better-value good.
3. **Placing the optimal bundle off the budget line.** The optimal bundle must exhaust the budget — it always lies *on* the budget line, not inside it.
4. **Getting the MRS direction wrong.** The MRS is MU_x/MU_y (the ratio of the good on the horizontal axis to the good on the vertical axis). Inverting this is one of the most common

algebraic errors in this topic.

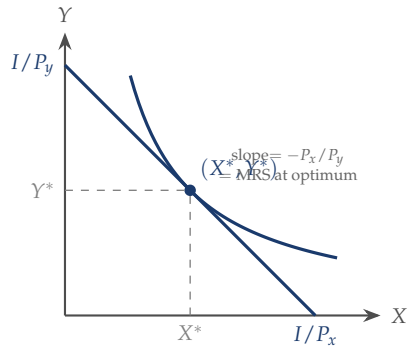
5. **Concluding that higher MU_x / P_x means spend less on X.** The opposite is true: if $MU_x / P_x > MU_y / P_y$, shift spending *toward* X, not away from it.

With the theory complete and the most common pitfalls flagged, the Chapter at a Glance on the next page gives you everything you need on a single revision sheet.

Chapter at a Glance

Your one-page revision guide — everything you need to review before an exam or tutorial.

The Consumer Optimum



Key Equations

Budget constraint	$P_x X + P_y Y = I$	p. 10
Slope of BC	$-P_x/P_y$	p. 10
Tangency condition	$\frac{P_x}{P_y} = \frac{MU_x}{MU_y}$	p. 17
Bang for the buck	$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$	p. 19
Cobb-Douglas demand	$X^* = \frac{\alpha}{\alpha+\beta}\frac{I}{P_x}$	p. 20

Indifference Curve Shapes at a Glance

Preference type	IC shape	MRS
Standard (both goods desired)	Downward-sloping, convex	Diminishing
Perfect substitutes	Straight line, slope $-a/b$	Constant $= a/b$
Perfect complements	L-shaped (kink on ratio ray)	0, ∞, or undefined
Neutral good (Y useless)	Vertical line	∞
Economic bad (Y bad)	Upward sloping	Negative

Five Key Terms

1. **IC** — bundles giving equal utility
2. **Budget constraint** — all affordable bundles
3. **MRS** — consumer's personal trade-off rate
4. **Tangency condition** — market rate = MRS at optimum
5. **Bang for the buck** — MU/P equalised at optimum

Which Method to Use?

- **Smooth convex IC** (e.g. Cobb-Douglas): use tangency: $P_x/P_y = MU_x/MU_y$
- **Perfect complements**: use kink condition: $X/a = Y/b$
- **Perfect substitutes**: compare MU_x/P_x vs MU_y/P_y; buy only the better-value good

Learning Objectives covered: LO1 (IC shapes) · LO2 (budget constraint) · LO3 (shifts and pivots) · LO4 (optimality conditions) · LO5 (bang for the buck) · LO6–LO7 (solving and selecting the right method).

2.9. Key Terms

Perfect Substitutes
Two goods that serve identical functions and can be traded for each other at a constant rate. Indifference curves are straight, parallel, downward-sloping lines.

Perfect Complements
Two goods that must be consumed together in fixed proportions. Extra units of one without the other provide no utility. Indifference curves are L-shaped (Leontief).

Neutral (Useless) Good
A good that provides no utility. Extra units leave total utility unchanged. Indifference curves are vertical lines.

Economic Bad
A good that confers negative utility (disutility). More of it reduces utility. Indifference curves are upward sloping.

Budget Constraint (BC)
The equation $I = P_x X + P_y Y$ describing all bundles the consumer can afford when spending exactly all income.

Budget Line
The graphical representation of the budget constraint: a downward-sloping line with Y-intercept I/P_y, X-intercept I/P_x, and slope $-P_x/P_y$.

Budget Set
The set of all affordable bundles — the region on and below the budget line.

Slope of the Budget Line
$-P_x/P_y$: the opportunity cost of good X in terms of good Y. For every extra unit of X bought, P_x/P_y units of Y must be given up.

Parallel Shift
What happens to the budget line when income changes (prices fixed): both intercepts move proportionally and the slope is unchanged.

Pivot (Rotation)
What happens to the budget line when a price changes (income and other price fixed): one intercept is fixed (the other good's axis), the other intercept moves, and the slope changes.

Optimal Consumption Bundle
The bundle (X^*, Y^*) that maximizes utility subject to the budget constraint. Found at the tangency of the budget line and the highest attainable indifference curve.

Tangency Condition
At the optimum: slope of BC = slope of IC, i.e. $P_x/P_y = MU_x/MU_y$.

Marginal Rate of Substitution (MRS)
The rate at which the consumer is willing to trade Y for X while staying at the same utility level. Equals MU_x/MU_y in magnitude. Also the negative slope of the indifference curve.

Bang for the Buck
The utility per dollar spent: MU_x/P_x for good X and MU_y/P_y for good Y. At the optimum, these are equalized: $MU_x/P_x = MU_y/P_y$.

Cobb-Douglas Utility
A utility function of the form $U(X, Y) = X^\alpha Y^\beta$ generating smooth, convex indifference curves. Demand satisfies $X^* = \frac{\alpha}{\alpha+\beta}\frac{I}{P_x}$, spending a fixed share of income on each good.

Supplementary Material: Video Resources

Recommended Videos

The following four videos have been selected for their clarity, popularity, and direct relevance to this chapter. Watch them in order for the best results.

1. Indifference Curves and the MRS *(Jacob Clifford, ACDC Economics)* ~9 min
https://www.youtube.com/watch?v=V7cSn87b3YE

Covers the definition and shape of indifference curves, why they slope downward and bow inward, and the marginal rate of substitution. One of the most-watched treatments on YouTube and closely aligned with Sections 2.1–2.6. Ideal as a visual complement before attempting the practice questions.

2. Budget Constraints *(Marginal Revolution University)* ~5 min
https://www.youtube.com/watch?v=pLhh_D5b_Lg

A concise, professionally produced explanation of the budget constraint equation, intercepts, slope, and affordable set. Produced by Tyler Cowen and Alex Tabarrok and consistently ranked among the best free economics content available. Directly reinforces Section 2.7–2.8.

3. Consumer Optimum: Putting It Together *(Khan Academy)* ~10 min
https://www.youtube.com/watch?v=uFHnFWMJE0A

Combines indifference curves and the budget constraint to locate the optimal bundle graphically and then algebraically. Walks through the tangency condition intuitively before introducing any algebra — a useful second pass on Sections 2.9–2.10, particularly for students who want to see the logic explained from scratch.

4. Cobb-Douglas Utility: Deriving Demand *(Economics in Many Lessons)* ~12 min
https://www.youtube.com/watch?v=7Qtr9AQBEPA

Derives the Cobb-Douglas demand functions step by step using the tangency condition, with numerical examples that closely mirror Worked Example 2 in Section 2.11. Recommended for students who want extra algebra practice before attempting the Level 3 questions.

2.10. Practice Questions

How to use these questions. Questions are arranged in three levels of difficulty. Level 1 builds vocabulary and geometric intuition; Level 2 requires algebra and diagram work; Level 3 demands synthesis. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Full solutions follow.

Level 1: Conceptual and Short-Answer

Q1. [★] Describe the shape of the indifference curve for each good type below and give a one-sentence economic justification.

(a) Two goods that are perfect substitutes.

(b) Two goods that are perfect complements.

(c) A neutral (useless) good paired with a normal good.

(d) An economic bad paired with a normal good.

Q2. [★] For each utility function, identify the preference type, describe the indifference curve shape, and state the MRS where defined.

(a) $U(X, Y) = 3X + 4Y$

(b) $U(X, Y) = \min(2X, Y)$

(c) $U(X, Y) = X$

(d) $U(X, Y) = X - Y^2$ (assume $Y \geq 0$)

Q3. [★] A consumer has $I = \$120$, $P_x = \$3$, $P_y = \$4$.

(a) Write the budget constraint equation.

(b) Find the X- and Y-intercepts.

(c) State the slope and explain its economic meaning.

(d) Sketch the budget line on a labelled diagram, shading the affordable set.

Q4. [★] Explain the difference between a parallel shift and a pivot of the budget line. What causes each? Give one real-world example of each.

Q5. [★] State the two conditions for an interior optimum. Explain in plain English why each is necessary — what goes wrong if either fails?

Q6. [★] Explain the "bang for the buck" condition in your own words. Give a concrete business example of what a manager should do when the condition is not satisfied.

Q7. [★] A consumer has $MU_x/P_x = 8$ and $MU_y/P_y = 5$ and her budget is exactly exhausted. Is she at an optimum? If not, how should she reallocate spending?

Q8. [★] A consumer says: "No matter how much coffee I drink, I always value the next unit of music streaming the same relative to coffee." What preferences is she describing? Write down a utility function and sketch her indifference map.

Q9. [★] Why will a rational consumer with positive marginal utilities never choose a bundle strictly inside the budget constraint? How does the answer change if one good is an economic bad?

Q10. [★] A student says: "Doubling my income always doubles my consumption of every good." Under what class of utility functions is this true? What is this property called?

Level 2: Intermediate Problems

Q11. [★★] A consumer has $I = \$80$, $P_x = \$2$, $P_y = \$4$.

(a) Draw the budget constraint with labelled intercepts.

(b) Income rises to \$120 (prices fixed). Draw the new line. Shift or pivot? Does slope change?

(c) Instead P_x rises to \$4 ($I = \$80$, P_y fixed). Draw. Shift or pivot?

(d) A classmate says the consumer in (c) is equally well off since income is unchanged. Is she correct? Explain using the budget set.

Q12. [★★] Tom and Jerry both have $I = \$60$, $P_x = \$3$, $P_y = \$6$. Tom: $U_T = 2X + Y$; Jerry: $U_J = \min(X, Y)$.

(a) Find Tom's optimal bundle. (Check MU_x/P_x vs MU_y/P_y.)

(b) Find Jerry's optimal bundle.

(c) Why do their solutions differ so dramatically? Link to indifference curve shapes.

Q13. [★★] $P_x = \$5$, $P_y = \$10$, $I = \$100$, $U = X^{1/2}Y^{1/2}$.

(a) Find (X^*, Y^*).

(b) Verify the tangency condition at the optimum.

(c) Verify budget exhaustion.

(d) P_y falls to \$5. Find the new optimum. What changed and why?

Q14. [★★] $U = \min(X, 2Y)$, $I = \$90$, $P_x = \$3$, $P_y = \$6$.

(a) Sketch two indifference curves. Where do the kinks lie?

(b) Find the optimal bundle.

(c) P_x rises to \$6. Find the new optimal bundle and describe the total effect on both goods.

Q15. [★★] $I = \$150$, $U = 4X + 3Y$.

(a) $P_x = \$4$, $P_y = \$3$: find the optimum.

(b) $P_x = \$3$, $P_y = \$4$: find the optimum.

(c) $P_x = \$4$, $P_y = \$4$: describe the solution set carefully.

(d) What do your answers illustrate about perfect substitutes?

Q16. [★★] (*Graphical.*) $I = \$60$, $P_x = \$3$, $P_y = \$2$, $U = X^{1/2}Y^{1/2}$.

(a) Find (X^*, Y^*).

(b) Mark an affordable but non-optimal bundle. Explain geometrically why it fails.

(c) Sketch the optimal indifference curve and confirm tangency.

Q17. [★★] (*Reverse problem.*) A consumer with $U = X^{1/2}Y^{1/2}$ and $I = \$120$ is observed at $(X^*, Y^*) = (30, 12)$.

(a) Use budget exhaustion to find P_y, given $P_x = \$2$.

(b) Verify your answer with the tangency condition.

(c) Repeat for the observed bundle $(20, 20)$ with the same income and $P_x = \$2$. Is this bundle consistent with optimisation?

Level 3: Advanced and Long-Answer Problems

Q18. [★★★] $I = \$200$, $P_x = \$4$, $P_y = \$8$, $U = X^{2/3}Y^{1/3}$.

(a) Derive the Marshallian demand functions $X^*(P_x, P_y, I)$ and $Y^*(P_x, P_y, I)$.

(b) Compute the optimum at the given parameters.

(c) What fraction of income is spent on each good? Is this consistent with the Cobb-Douglas result?

(d) Income rises to \$300. Find the new optimum and state the percentage change in each good. Are X and Y normal goods?

Q19. [★★★] Weekly budget \$60: coffee (X, \$3), textbooks (Y, \$15). $U = X^{3/4}Y^{1/4}$.

(a) Find the optimal bundle and compute utility at the optimum.

(b) Textbooks go on sale at \$10. Find the new optimal bundle.

(c) Draw both budget lines on one diagram, marking both optima. Shift or pivot? Which intercept is fixed?

(d) What percentage does textbook consumption change? Is this consistent with Cobb-Douglas spending shares?

Q20. [★★★] $U(X, Y) = \ln X + 2 \ln Y$.

(a) Derive the MRS. (*Note*: this is a monotone transformation of XY^2.)

(b) $I = \$120$, $P_x = \$2$, $P_y = \$4$: find the optimal bundle.

(c) What fraction of income is spent on each good?

(d) Derive the general demand functions $X^*(P_x, P_y, I)$ and $Y^*(P_x, P_y, I)$.

Q21. [★★★] (True/False — justify carefully.)

(a) A rise in P_x rotates the budget line inward around the Y-intercept.

(b) For perfect complements, the optimal bundle is always at the kink.

(c) At the optimum, slope of IC $= -P_x/P_y$ for *all* preference types.

(d) If $MU_x/P_x > MU_y/P_y$, the consumer should shift spending toward Y.

(e) Doubling all prices has the same effect on the budget set as halving income.

(f) For $U = aX + bY$, the consumer always buys only the good with the higher utility-per-dollar ratio.

Q22. [★★★] $I = \$240$, $P_x = \$6$ (food), $P_y = \$12$ (housing), $U = X^{1/2}Y^{1/2}$.

(a) Find the optimal bundle.

(b) P_y rises to \$16. Find the new optimal bundle.

(c) Draw both budget lines and mark both optima on a labelled diagram.

(d) How much must income rise (with $P_y = \$16$) to restore the original utility level?

Q23. [★★★] $U = XY + X$, $I = \$100$, $P_x = \$5$, $P_y = \$4$.

(a) Find MU_x, MU_y, and the MRS.

(b) Find the optimal bundle. Show all steps.

(c) Verify both optimality conditions.

(d) How does the $+X$ term affect the MRS relative to a standard Cobb-Douglas? What does it imply about the consumer's willingness to trade?

Q24. [★★★] $U = X^{1/3}Y^{2/3}$, $I = \$90$, $P_x = \$3$, $P_y = \$3$.

(a) Find the optimal bundle.

(b) P_y rises to \$6. Find the new optimum.

(c) Sketch both budget lines and both optima.

(d) Income rises to \$180 ($P_x = \$3$, $P_y = \$6$). Find the new optimum and the income effect.

(e) Confirm both goods are normal using the demand functions.

Q25. [★★★] $I = \$100$, $U = \min(X, Y)$, $P_x = \$3$, $P_y = \$7$.

(a) Find the optimal bundle.

(b) Why does the tangency condition not apply? State the correct condition.

(c) Both prices double. Find the new optimum and state the general principle illustrated.

(d) Show algebraically that $X^* = Y^*$ always holds for $U = \min(X, Y)$.

Q26. [★★★] I given, $P_x = P_y = P$, $U = X^{1/2}Y^{1/2}$.

(a) Find the optimal bundle in terms of I and P.

(b) Both prices double to $2P$. How does the optimum change?

(c) Compare doubling all prices to halving income. What principle does this illustrate?

(d) P_x doubles but P_y stays at P. How does this differ from (b)? Which change hurts more, and why?

Q27. [★★★] (*Firm application.*) A firm uses labour L and capital K with production function $Q = L^{1/2}K^{1/2}$, wage $w = \$10$, rental rate $r = \$20$, and cost budget $C = \$200$.

(a) Draw the isocost line. State its intercepts and slope.

(b) Write the two conditions for cost-minimising input choice (analogous to consumer optimality).

(c) Solve for the cost-minimising (L^*, K^*) to produce $Q = 10$ units.

(d) What structural parallels exist between this problem and consumer optimisation? What plays the role of the budget constraint? Of the indifference curve?

Full solutions follow. Attempt all questions independently before consulting the solutions.

Solutions to Practice Questions

Note on numbering. Q1–Q10 are Level 1; Q11–Q17 are Level 2; Q18–Q27 are Level 3. The old Q22–Q26 from previous editions correspond to Q8–Q10 and Q16–Q17 in this edition.

Solution to Q1

(a) **Perfect substitutes:** straight, parallel, downward-sloping lines. The consumer is always willing to trade the goods at a constant ratio (constant MRS), so the IC has constant slope.

(b) **Perfect complements:** L-shaped. Extra units of one good without the other provide zero marginal utility, so horizontal and vertical arms represent equal utility.

(c) **Neutral good (Y):** vertical straight lines. Utility only depends on X; any amount of Y at fixed X gives the same utility. Increasing utility means moving right.

(d) **Economic bad (Y):** upward sloping. More Y reduces utility, so to stay on the same IC, the consumer needs compensation in the form of more X. The IC therefore rises as Y increases.

Solution to Q2

(a) $U = 3X + 4Y$: linear utility $\Rightarrow$ perfect substitutes with constant MRS $= 3/4$. ICs are parallel straight lines with slope $-3/4$.

(b) $U = \min(2X, Y)$: perfect complements with kinks on the ray $2X = Y$ (i.e., $Y = 2X$). L-shaped ICs with horizontal arm at $Y = 2X_0$ and vertical arm at $X = X_0$.

(c) $U = X$: utility depends only on $X \Rightarrow Y$ is a neutral good. ICs are vertical lines (constant X, any Y).

(d) $U = X - Y^2$: $MU_y = -2Y < 0$ for $Y > 0 \Rightarrow Y$ is an economic bad. ICs slope upward: to compensate for more Y, the consumer needs more X.

Solution to Q3

(a) $3X + 4Y = 120$.

(b) $X_{\max} = 120/3 = 40$; $Y_{\max} = 120/4 = 30$.

(c) Slope $= -3/4$.

(d) Draw axes (Y vertical, X horizontal). Mark $(0, 30)$ on Y-axis and $(40, 0)$ on X-axis. Connect with a straight line labeled $3X + 4Y = 120$. Shade the affordable region below the line.

Solution to Q4

Parallel shift: both intercepts change proportionally, slope unchanged. Caused by an income change (prices fixed). Income up $\Rightarrow$ outward parallel shift; income down $\Rightarrow$ inward.

Pivot (rotation): one intercept is fixed (the other good's axis), the second intercept changes,

and the slope changes. Caused by a price change for one good (income and other price fixed). P_x falls $\Rightarrow$ X-intercept moves out, Y-intercept fixed; line becomes flatter.

Solution to Q5

Condition 1 (budget exhausted): $P_x X^* + P_y Y^* = I$ (see Equation 1 in the notes). This is necessary because both goods are desirable: any income not spent is wasteful — the consumer could increase utility by buying more of either good.

Condition 2 (tangency): slope of BC = slope of IC, i.e. $P_x/P_y = MU_x/MU_y$ (see Equation 3 in the notes). This is derived by setting the slope of the budget line equal to the MRS: $-P_x/P_y = -MU_x/MU_y$, then cancelling the negatives. It is necessary because if the slopes differ, the IC crosses the budget line, meaning there is an affordable bundle on a higher IC. Only at tangency is it impossible to reach a higher IC within the budget.

Solution to Q6

The bang-for-the-buck condition (Equation 4 in the notes) states $MU_x/P_x = MU_y/P_y$. It is derived from the tangency condition (Equation 3) by rearranging $P_x/P_y = MU_x/MU_y$ to give $MU_x/P_x = MU_y/P_y$. At the optimum, the extra utility gained per dollar spent is the same for every good. If this were not true — say $MU_x/P_x > MU_y/P_y$ — the consumer gets more utility per dollar from X than from Y. They should reduce Y by \$1 (losing MU_y/P_y utility) and increase X by \$1 (gaining MU_x/P_x utility). Since the gain exceeds the loss, total utility increases. This reallocation should continue until the ratios are equalized.

Example: If tickets to a concert give 20 utils per dollar and coffee gives 10 utils per dollar, the consumer should shift spending from coffee to concert tickets until the ratios equalize (as diminishing MU sets in).

Solution to Q7

The consumer is not at an optimum. Since $MU_x/P_x = 8 > 5 = MU_y/P_y$, good X provides more utility per dollar than good Y. The consumer should **increase** spending on X and decrease spending on Y. As they do so, diminishing marginal utility will lower MU_x and raise MU_y (assuming standard convex preferences), eventually equating the two ratios.

Solution to Q8

(a) $2X + 4Y = 80$. $X_{\text{max}} = 40$; $Y_{\text{max}} = 20$. Slope $= -1/2$.

(b) $2X + 4Y = 120$. $X_{\text{max}} = 60$; $Y_{\text{max}} = 30$. Parallel outward shift. Slope is still $-1/2$ — unchanged.

(c) $4X + 4Y = 80$. $X_{\text{max}} = 20$; $Y_{\text{max}} = 20$ (unchanged). Pivot inward around $Y_{\text{max}} = 20$. Slope changes from $-1/2$ to -1 (steeper).

(d) The classmate is incorrect. Although nominal income is unchanged, the budget set is smaller in (c): the new budget line lies inside the old one for $X > 0$. The consumer has fewer affordable options. A higher P_x means every unit of X costs more, effectively reducing the consumer's purchasing power.

Solution to Q9

(a) $U_T = 2X + Y$. Check bang for the buck: $MU_x/P_x = 2/3$; $MU_y/P_y = 1/6$. Since $2/3 > 1/6$, X gives more utility per dollar. Tom spends all income on X: $X^* = 60/3 = 20$, $Y^* = 0$.

(b) $U_J = \min(X, Y)$. Optimum at kink: $X = Y$. Budget: $3X + 6X = 60 \Rightarrow 9X = 60 \Rightarrow X^* = 20/3 \approx 6.67$; $Y^* = 20/3$.

(c) Tom views the goods as perfect substitutes (linear utility) and always chooses the one with more utility per dollar, leading to a corner solution. Jerry treats them as perfect complements and always consumes equal amounts, leading to an interior kink solution.

Solution to Q10

$U = X^{1/2}Y^{1/2}$; $I = 100$; $P_x = 5$; $P_y = 10$.

(a) Tangency: $Y/X = P_x/P_y = 1/2 \Rightarrow Y = X/2$. Budget: $5X + 10(X/2) = 100 \Rightarrow 10X = 100 \Rightarrow X^* = 10$; $Y^* = 5$.

(b) MRS at $(10, 5)$: $-Y/X = -5/10 = -1/2$. Slope of BC: $-P_x/P_y = -5/10 = -1/2$. ✓

(c) $5(10) + 10(5) = 50 + 50 = 100$. ✓

(d) $P_y' = 5$: Tangency: $Y/X = 5/5 = 1 \Rightarrow Y = X$. Budget: $5X + 5X = 100 \Rightarrow X^{**} = 10$; $Y^{**} = 10$. X is unchanged; Y doubled. This reflects the equal-expenditure-share property of $\sqrt{XY}$.

Solution to Q11

$U = \min(X, 2Y)$; $I = 90$; $P_x = 3$; $P_y = 6$.

(a) ICs are L-shaped with kinks on the ray $X = 2Y$. On this ray both arguments of min are equal. Example kinks: $(2, 1)$, $(4, 2)$, $(6, 3)$ etc.

(b) Optimum at kink: $X = 2Y$. Budget: $3(2Y) + 6Y = 90 \Rightarrow 12Y = 90 \Rightarrow Y^* = 7.5$; $X^* = 15$.

(c) $P_x' = 6$: $X = 2Y$ still (ratio unchanged). Budget: $6(2Y) + 6Y = 90 \Rightarrow 18Y = 90 \Rightarrow Y^{**} = 5$; $X^{**} = 10$. Total effect: X falls from 15 to 10; Y falls from 7.5 to 5.

Solution to Q12

$U = 4X + 3Y$; $I = 150$. MU_x/P_x vs MU_y/P_y determines the corner.

(a) $P_x = 4$, $P_y = 3$: $MU_x/P_x = 4/4 = 1$; $MU_y/P_y = 3/3 = 1$. Equal — **any** bundle on the budget line is optimal (the budget line overlaps an IC). The consumer is indifferent among all affordable combinations.

(b) $P_x = 3$, $P_y = 4$: $MU_x/P_x = 4/3 > 3/4 = MU_y/P_y$. Spend all on X: $X^* = 150/3 = 50$, $Y^* = 0$.

(c) $P_x = 4$, $P_y = 4$: $MU_x/P_x = 4/4 = 1 > 3/4 = MU_y/P_y$. Spend all on X: $X^* = 150/4 = 37.5$, $Y^* = 0$.

(d) Perfect substitutes generate corner solutions (spend all on the good with the higher utility per dollar). When utility per dollar is equal, any point on the budget line is optimal.

Solution to Q13

$U = X^{2/3}Y^{1/3}$; $\alpha = 2/3$, $\beta = 1/3$.

(a) Using the general Cobb-Douglas demand formula derived in Section 2.8 (Worked Example 2): $X^* = \frac{\alpha}{\alpha+\beta} \cdot \frac{I}{P_x} = \frac{2I}{3P_x}$; $Y^* = \frac{\beta}{\alpha+\beta} \cdot \frac{I}{P_y} = \frac{I}{3P_y}$. The derivation applies the tangency condition (Eq. 3) and budget exhaustion (Eq. 1) simultaneously.

(b) $X^* = 2(200)/(3 \cdot 4) = 100/3 \approx 33.3$; $Y^* = 200/(3 \cdot 8) = 25/3 \approx 8.33$.

(c) Spending on X: $4 \times 100/3 \approx \$133.3 = 2/3 \times 200$. Spending on Y: $8 \times 25/3 \approx \$66.7 = 1/3 \times 200$. Consistent with Cobb-Douglas: share on X is $\alpha/(\alpha+\beta) = 2/3$ and on Y is $1/3$.

(d) $I' = 300$: $X^{**} = 2(300)/(12) = 50$; $Y^{**} = 300/24 = 12.5$. Both rose by 50%. Both are normal goods.

Solution to Q14

$U = X^{3/4}Y^{1/4}$; $I = 60$; $P_x = 3$; $P_y = 15$.

(a) $X^* = \frac{3/4}{1} \cdot \frac{60}{3} = 15$; $Y^* = \frac{1/4}{1} \cdot \frac{60}{15} = 1$.

(b) $U^* = 15^{3/4} \cdot 1^{1/4} = 15^{3/4} \approx 8.09$.

(c) $P_y' = 10$: $X^{**} = (3/4)(60/3) = 15$ (unchanged); $Y^{**} = (1/4)(60/10) = 1.5$. Textbook demand rises from 1 to 1.5.

(d) *Diagram:* Y-axis = textbooks, X-axis = coffee. Old BC: $3X + 15Y = 60$ with X-int= 20 and Y-int= 4. New BC: $3X + 10Y = 60$ with X-int= 20 (unchanged) and Y-int= 6. BC **pivots** around X-int= 20 (since P_x is unchanged). Old optimum: $(15, 1)$; new optimum: $(15, 1.5)$.

Solution to Q15

$U = \ln X + 2 \ln Y$. Note: $\ln X + 2 \ln Y = \ln(X \cdot Y^2)$, same as Cobb-Douglas X^1Y^2 (monotone transformation). So $\alpha = 1$, $\beta = 2$.

(a) $MU_x = 1/X$; $MU_y = 2/Y$. MRS $= (1/X)/(2/Y) = Y/(2X)$ in magnitude. So slope of IC $= -Y/(2X)$.

(b) Tangency: $Y/(2X) = P_x/P_y = 2/4 = 1/2 \Rightarrow Y/(2X) = 1/2 \Rightarrow Y = X$. Budget: $2X + 4X = 120 \Rightarrow 6X = 120 \Rightarrow X^* = 20$; $Y^* = 20$.

(c) Spending on X: $2 \times 20 = \$40 = 1/3 \times 120$. On Y: $4 \times 20 = \$80 = 2/3 \times 120$. Shares:

$\alpha/(\alpha+\beta) = 1/3$ on X and $2/3$ on Y. ✓

(d) $X^* = \frac{1}{3} \cdot \frac{I}{P_x}$; $Y^* = \frac{2}{3} \cdot \frac{I}{P_y}$.

Solution to Q16

(a) **True.** When P_x rises, $X_{\max} = I/P_x$ falls (inward pivot) while $Y_{\max} = I/P_y$ is unchanged. The line rotates inward around the Y-intercept.

(b) **True.** For perfect complements, extra units of one good are useless without the other. The consumer always chooses the kink (where the two goods are in the correct ratio), since moving away from the kink along either arm provides no extra utility.

(c) **Uncertain (False for corner solutions).** The tangency condition $-P_x/P_y =$ slope of IC holds for interior optima with well-behaved (smooth, convex) ICs. For perfect substitutes (linear IC) or perfect complements (L-shaped IC), the optimum is typically at a corner or kink, and tangency does not characterize the solution.

(d) **False.** If $MU_x/P_x > MU_y/P_y$, good X gives more utility per dollar. The consumer should **increase** spending on X and decrease spending on Y, not the other way around.

(e) **True.** If all prices double, the budget constraint becomes $2P_xX + 2P_yY = I$, equivalent to $P_xX + P_yY = I/2$. Both intercepts halve, exactly as if income were halved. The budget set is identical.

(f) **True (generally).** For $U = aX + bY$, $MU_x/P_x = a/P_x$ and $MU_y/P_y = b/P_y$. The consumer spends all income on whichever good has the higher ratio. Only if $a/P_x = b/P_y$ is the consumer indifferent among all bundles on the budget line.

Solution to Q17

$U = X^{1/2}Y^{1/2}$; $I = 240$; $P_x = 6$; $P_y = 12$.

(a) $X^* = 240/(2 \cdot 6) = 20$; $Y^* = 240/(2 \cdot 12) = 10$.

(b) $P_y' = 16$: $X^{**} = 240/(2 \cdot 6) = 20$; $Y^{**} = 240/(2 \cdot 16) = 7.5$.

(c) *Diagram:* Y-axis (housing), X-axis (food). Old BC: $6X + 12Y = 240$ (X-int$=$ 40, Y-int$=$ 20). New BC: $6X + 16Y = 240$ (X-int$=$ 40, Y-int$=$ 15) — pivot around X-int$=$ 40. Mark $(20, 10)$ and $(20, 7.5)$.

(d) Utility at (a): $U_1 = \sqrt{20 \cdot 10} = \sqrt{200}$. At new prices $P_y = 16$, need (X, Y) with $\sqrt{XY} = \sqrt{200}$ and $6X + 16Y = I'$. At optimum with these prices: $X/Y = P_y/P_x = 16/6 = 8/3$, so $X = (8/3)Y$. Then $XY = (8/3)Y^2 = 200 \Rightarrow Y^2 = 75 \Rightarrow Y = 5\sqrt{3}$; $X = (40\sqrt{3})/3$. Income needed: $I' = 6 \cdot (40\sqrt{3}/3) + 16 \cdot 5\sqrt{3} = 80\sqrt{3} + 80\sqrt{3} = 160\sqrt{3} \approx \$277.1$.

Solution to Q18

$U = XY + X$.

(a) $MU_x = Y + 1$; $MU_y = X$.

(b) MRS $= (Y+1)/X$ in magnitude.

(c) Tangency: $(Y+1)/X = P_x/P_y = 5/4 \Rightarrow 4(Y+1) = 5X \Rightarrow Y = (5X-4)/4$. Budget: $5X + 4Y = 100 \Rightarrow 5X + 5X - 4 = 100 \Rightarrow 10X = 104 \Rightarrow X^* = 10.4$; $Y^* = (5 \times 10.4 - 4)/4 = (52-4)/4 = 48/4 = 12$.

(d) Budget: $5(10.4) + 4(12) = 52 + 48 = 100$. ✓MRS at $(10.4, 12)$: $(12+1)/10.4 = 13/10.4 = 1.25 = 5/4 = P_x/P_y$. ✓

Solution to Q19

$U = X^{1/2}Y^{1/2}$; $I = 60$; $P_x = 2$; $P_y = 3$.

(a) $X^* = 60/(2 \cdot 2) = 15$; $Y^* = 60/(2 \cdot 3) = 10$.

(b) Bundle $(6, 12)$: $2(6) + 3(12) = 12 + 36 = 48 < 60$. **Affordable** but not optimal (inside BC). Utility: $\sqrt{72} < \sqrt{150} = U^*$. Consumer can improve by spending remaining \$12 on either good (or both).

(c) Bundle $(15, 10)$: $2(15) + 3(10) = 30 + 30 = 60$. **Affordable** (on BC). Utility: $\sqrt{150} = U^*$. **This is optimal.** MRS at $(15, 10)$: $-Y/X = -10/15 = -2/3 = -P_x/P_y$. ✓

(d) *Diagram:* Y-axis (Qty Y), X-axis (Qty X). Budget line from $(0, 20)$ to $(30, 0)$. Optimal IC tangent at $(15, 10)$. Bundle $(6, 12)$ inside the budget set; $(15, 10)$ on the BC at the tangency.

Solution to Q20

$U = X^{1/3}Y^{2/3}$; $\alpha = 1/3$, $\beta = 2/3$.

(a) $X^* = (1/3)(90/3) = 10$; $Y^* = (2/3)(90/3) = 20$.

(b) $P_y = 6$: $X^{**} = (1/3)(90/3) = 10$ (unchanged); $Y^{**} = (2/3)(90/6) = 10$.

(c) *Diagram:* Y-axis, X-axis. Old BC: $3X + 3Y = 90$ (X-int$= 30$, Y-int$= 30$); new BC: $3X + 6Y = 90$ (X-int$= 30$, Y-int$= 15$). Pivot around X-int$= 30$. Old optimum: $(10, 20)$; new: $(10, 10)$.

(d) $I' = 180$, $P_y = 6$: $X^{***} = (1/3)(180/3) = 20$; $Y^{***} = (2/3)(180/6) = 20$. Income effect: $\Delta X = 20 - 10 = +10$; $\Delta Y = 20 - 10 = +10$. Both increase.

(e) From demand functions: $\partial X^*/\partial I = 1/(3P_x) > 0$ and $\partial Y^*/\partial I = 2/(3P_y) > 0$. Both normal goods.

Solution to Q21

$U = \min(X, Y)$; $I = 100$; $P_x = 3$; $P_y = 7$.

(a) Optimum at kink: $X = Y$. Budget: $3X + 7X = 100 \Rightarrow 10X = 100 \Rightarrow X^* = 10$; $Y^* = 10$.

(b) The tangency condition applies to smooth IC-BC tangencies. Here the IC is L-shaped and the optimum is at the kink, where the slope of the IC is undefined (it has a cor-

ner). The condition MRS $= P_x/P_y$ does not apply because MRS is not defined at the kink. Instead, the kink condition $X = Y$ plus the budget constraint characterizes the optimum.

(c) $P'_x = 6$, $P'_y = 14$: kink: $X = Y$. Budget: $6X + 14X = 100 \Rightarrow 20X = 100 \Rightarrow X^{**} = 5$; $Y^{**} = 5$. Quantities halved, consistent with the effect of proportional price doubling (equivalent to halving income).

(d) For $U = \min(X, Y)$: if $X > Y$, extra X is wasted. If $X < Y$, extra Y is wasted. Utility only increases by increasing both simultaneously. So the consumer always chooses $X = Y$. Combined with the budget $P_xX + P_yY = I$: $X^* = Y^* = I/(P_x + P_y)$.

Solution to Q22

The consumer is describing **perfect substitutes**. Her marginal rate of substitution between music streaming and coffee is constant — she always values one additional unit of streaming relative to coffee at the same fixed rate, regardless of how much of either she already has. Her indifference map consists of straight, parallel, downward-sloping lines. The slope of each IC equals the (negative of the) constant trade-off rate she described.

Solution to Q23

A rational consumer will never choose a bundle strictly inside the budget constraint because at any such point, she has unspent income. Since both goods are desirable (positive marginal utility), she can increase utility by spending the remaining income on more of either X or Y, reaching a higher indifference curve. The optimal bundle must lie on the budget line — all income is exhausted.

If one good is a "bad" (negative MU), the consumer would actually want to consume zero of that bad. In that case, she might optimally spend all income on the good and zero on the bad — a corner solution on the budget line, not interior.

Solution to Q24

This property holds for **Cobb-Douglas** utility functions $U(X, Y) = X^\alpha Y^\beta$. The demand functions are $X^* = \frac{\alpha}{\alpha+\beta} \cdot \frac{I}{P_x}$ and $Y^* = \frac{\beta}{\alpha+\beta} \cdot \frac{I}{P_y}$, which are both linear in income. Doubling I exactly doubles X^* and Y^*. This property is sometimes called **homotheticity**: preferences are homothetic when optimal demands are proportional to income. More generally, any homothetic utility function shares this property.

Solution to Q25

Setup: With $I = \$60$, $P_x = \$3$, $P_y = \$2$: $X_{\max} = 20$, $Y_{\max} = 30$, slope $= -3/2$.

For $U = X^{1/2}Y^{1/2}$, the optimal bundle is $X^* = 60/(2 \cdot 3) = 10$ and $Y^* = 60/(2 \cdot 2) = 15$.

A non-optimal but affordable bundle might be $(5, 10)$: this is inside the BC (cost $= 15 + 20 = 35 < 60$) and lies on a lower IC. Moving to the budget line and then to the tangency at $(10, 15)$ reaches a higher IC.

The optimal IC at $(10, 15)$ is tangent to the budget line: no part of that IC lies below and

to the right of the budget line, meaning the consumer cannot reach a higher IC without violating the budget. This is what makes it the optimum.

Solution to Q26

$U = X^{1/2}Y^{1/2}$; $P_x = P_y = P$.

(a) Using the Cobb-Douglas formula: $X^* = \frac{I}{2P}$ and $Y^* = \frac{I}{2P}$. Equal prices give equal quantities.

(b) With $P_x = P_y = 2P$: $X^{**} = \frac{I}{4P}$ and $Y^{**} = \frac{I}{4P}$. Demand for each good halves.

(c) **Principle:** Doubling all prices is equivalent to halving income — real purchasing power falls by half even though nominal income is unchanged.

(d) If only P_x doubles (to $2P$, P_y stays at P): tangency gives $Y/X = 2P/P = 2$, so $Y = 2X$. Budget: $2PX + P(2X) = I \Rightarrow X^* = I/(4P)$, $Y^* = I/(2P)$. The consumer substitutes away from the now-more-expensive X into Y. This hurts more than a proportional price increase because the pivot changes relative prices, distorting consumption away from the previously preferred mix. The consumer ends up on a lower IC than in part (b) despite spending the same income.

Solution to Q17 (Reverse Problem)

$U = X^{1/2}Y^{1/2}$, $I = \$120$, $P_x = \$2$.

(a) Observed bundle $(30, 12)$. Budget exhaustion: $2(30) + P_y(12) = 120 \Rightarrow 12P_y = 60 \Rightarrow P_y = \$5$.

(b) Tangency condition: MRS $= Y/X = 12/30 = 2/5$. Price ratio: $P_x/P_y = 2/5$. ✓ Consistent with optimisation.

(c) Observed bundle $(20, 20)$: budget exhaustion gives $2(20) + P_y(20) = 120 \Rightarrow P_y = \$4$. Tangency check: MRS $= Y/X = 20/20 = 1$. Price ratio: $2/4 = 1/2 \neq 1$. This bundle fails the tangency condition — it is on the budget line but is *not* a utility maximum at these prices.

Solution to Q27 (Firm Application)

$Q = L^{1/2}K^{1/2}$, $w = \$10$, $r = \$20$, $C = \$200$.

(a) Isocost line: $10L + 20K = 200$. L-intercept: $L = 20$; K-intercept: $K = 10$. Slope: $-w/r = -10/20 = -1/2$.

(b) The two optimality conditions mirror the consumer problem exactly:

- *Cost exhaustion*: $wL^* + rK^* = C$ (spend the full cost budget).
- *Tangency*: $MP_L/MP_K = w/r$ (the technical rate of substitution equals the input price ratio).

(c) $MP_L = \frac{1}{2}L^{-1/2}K^{1/2}$; $MP_K = \frac{1}{2}L^{1/2}K^{-1/2}$. Tangency: $K/L = w/r = 1/2$, so $K = L/2$.

Target output: $Q = L^{1/2}(L/2)^{1/2} = L/\sqrt{2} = 10 \Rightarrow L^* = 10\sqrt{2} \approx 14.14$; $K^* = 5\sqrt{2} \approx 7.07$.
Verify cost: $10(10\sqrt{2}) + 20(5\sqrt{2}) = 100\sqrt{2} + 100\sqrt{2} = 200\sqrt{2} \approx \$283$. Since $\$283 > \$200$, the firm cannot produce 10 units on this budget. The cost-minimising bundle at $C = \$200$ produces $Q = (200/(2\sqrt{w \cdot r})) = 200/(2\sqrt{200}) \approx 7.07$ units.

(d) **Structural parallels**: The isocost line ($wL + rK = C$) is the exact analogue of the budget constraint ($P_x X + P_y Y = I$). The isoquant ($Q = \bar{Q}$) plays the role of the indifference curve. The firm minimises cost subject to a production target, just as the consumer maximises utility subject to a budget. The tangency condition $MP_L/MP_K = w/r$ mirrors $MU_x/MU_y = P_x/P_y$. This is why mastering consumer theory now pays dividends when you encounter production theory in Chapter 6.

End of Week 2 Notes